

CredibleX Growth Intelligence Engine

Project Charter · AI-Powered SME Financing & Partner Opportunity Intelligence

Type	Independent portfolio prototype
Stack	Python · SQL · Excel · HTML / CSS / JavaScript · GenAI
Status	Concept-to-prototype, synthetic data

Goal

Build an independent prototype showing how CredibleX could use data and AI to identify financing opportunities, match SMEs with suitable financing products, and prioritise potential ecosystem partners.

Business Problem

How can data and AI help identify SMEs with potential financing needs and uncover high-potential embedded-finance partnerships?

CredibleX operates as an embedded-finance lender, distributing working-capital products through platform partners rather than direct SME acquisition (see *CredibleX_Research.md*). That model creates two linked intelligence problems this project addresses together: which SMEs represent the strongest financing opportunity and with which product, and which potential partners offer the best distribution reach into that opportunity.

Objectives

- Identify SMEs with strong financing opportunities.
- Recommend a suitable financing product for each SME.
- Identify and prioritise potential ecosystem partners.
- Use AI to explain the analytical results in business language.
- Present everything through an interactive HTML dashboard.

Scope

Included	SME analytics, financing-opportunity scoring, product matching, partner intelligence, AI-generated explanations, an early-warning classification layer.
----------	---

Excluded	Actual credit approval, real customer data, and CredibleX's real underwriting methodology. All scores in this prototype are commercial-opportunity heuristics, not credit-risk models.
----------	--

Deliverables

- Project Charter (this document) and a short research brief on CredibleX and embedded finance.
- Two labelled synthetic datasets: 1,000 SMEs and 50 potential partners.

- A Python + SQL analytical engine (three notebooks + one SQL script) covering SME opportunity scoring, product recommendation, partner opportunity scoring, and early-warning classification.
- An AI explanation layer that turns calculated results into AI Financing Briefs, AI Partner Briefs, and an AI Executive Summary.
- A four-page interactive HTML/CSS/JS dashboard (Executive Overview, SME Intelligence, Partner Intelligence, AI Insights), deployable on GitHub Pages.
- A final report summarising methodology, findings, and business impact.

Success Criteria

- The dashboard runs standalone (no backend) and is shareable as a single GitHub Pages link.
- Every score and recommendation shown in the dashboard traces back to a deterministic, documented calculation in the notebooks or SQL script.
- The AI layer explains results without inventing figures, verified by spot-checking briefs against their source rows.
- The repository is small, clearly organised, and readable by a technical reviewer in minutes.

Key Risks & Mitigations

Risk	Mitigation
No access to real CredibleX data	Clearly labelled synthetic datasets, grounded in public research on CredibleX’s real product suite and partner model.
Scores could read as a credit/risk model	Explicitly framed and labelled as a commercial-opportunity score throughout; early-warning kept separate and simple.
AI hallucination in generated briefs	Prompts inject only pre-computed fields (see ai/prompts.md); AI explains, never calculates.

Independent prototype built for portfolio purposes. Not affiliated with, commissioned by, or endorsed by CredibleX. All SME and partner data referenced in this project is synthetically generated and does not represent real CredibleX customers or partners.